Analysis of the Mental Rotation 2D experiment

Volunteers took the experiment at https://github.com/chrplr/goxpyriment/tree/main/examples/Mental-Rotation-2D

```
library(dplyr)
library(ggplot2)
set.seed(42)  # makes the permutation tests reproducible
```

Importing data into a single data frame

```
files <- list.files("./data", pattern = "\\.csv$", full.names=T)
print(length(files))
```

```
## [1] 16
```

```
files
```

```
##  [1] "./data/Mental-Rotation-2D_sub-000_date-20260924-094820.csv"
##  [2] "./data/Mental-Rotation-2D_sub-000_date-20260924-095043.csv"
##  [3] "./data/Mental-Rotation-2D_sub-000_date-20260924-095219.csv"
##  [4] "./data/Mental-Rotation-2D_sub-001_date-20260924-064330.csv"
##  [5] "./data/Mental-Rotation-2D_sub-001_date-20260924-095036.csv"
##  [6] "./data/Mental-Rotation-2D_sub-001_date-20260924-095059.csv"
##  [7] "./data/Mental-Rotation-2D_sub-001_date-20260924-095218.csv"
##  [8] "./data/Mental-Rotation-2D_sub-001_date-20260924-095332.csv"
##  [9] "./data/Mental-Rotation-2D_sub-001_date-20260924-101256.csv"
## [10] "./data/Mental-Rotation-2D_sub-001_date-20260926-211937.csv"
## [11] "./data/Mental-Rotation-2D_sub-001_date-20260927-161502.csv"
## [12] "./data/Mental-Rotation-2D_sub-001_date-20260927-181811.csv"
## [13] "./data/Mental-Rotation-2D_sub-020_date-20260926-162037.csv"
## [14] "./data/Mental-Rotation-2D_sub-1612_date-20260930-140849.csv"
## [15] "./data/res.csv"
## [16] "./data/res2.csv"
```

```
read_all <- function(files) {
  bind_rows(lapply(files, function(f) {
    d <- read.csv(f)
    d$file_name <- f
    # the csv files store "true"/"false" in lowercase, which R does not parse as logical
    d$is_correct <- tolower(as.character(d$is_correct)) == "true"
    d
  })) %>%
    relocate(file_name)
}

data <- read_all(files)
head(data)
```

```
##                                                file_name subject_id
```

```
## 1 ./data/Mental-Rotation-2D_sub-000_date-20260924-094820.csv          0
## 2 ./data/Mental-Rotation-2D_sub-000_date-20260924-094820.csv          0
## 3 ./data/Mental-Rotation-2D_sub-000_date-20260924-094820.csv          0
## 4 ./data/Mental-Rotation-2D_sub-000_date-20260924-094820.csv          0
## 5 ./data/Mental-Rotation-2D_sub-000_date-20260924-094820.csv          0
## 6 ./data/Mental-Rotation-2D_sub-000_date-20260924-094820.csv          0
##   trial_idx angle condition response is_correct   rt
## 1         1    40  mirrored mirrored       TRUE 1649
## 2         2   120      same     same       TRUE 1433
## 3         3   160      same     same       TRUE 1168
## 4         4    80      same     same       TRUE  906
## 5         5    80  mirrored mirrored       TRUE  930
## 6         6   120      same     same       TRUE  800
```

Data inspection and cleaning

Number of datapoints in each file

```
data %>% count(file_name)
```

```
##                                                  file_name  n
## 1   ./data/Mental-Rotation-2D_sub-000_date-20260924-094820.csv 40
## 2   ./data/Mental-Rotation-2D_sub-000_date-20260924-095043.csv 40
## 3   ./data/Mental-Rotation-2D_sub-000_date-20260924-095219.csv 40
## 4   ./data/Mental-Rotation-2D_sub-001_date-20260924-064330.csv 13
## 5   ./data/Mental-Rotation-2D_sub-001_date-20260924-095036.csv 40
## 6   ./data/Mental-Rotation-2D_sub-001_date-20260924-095059.csv 40
## 7   ./data/Mental-Rotation-2D_sub-001_date-20260924-095218.csv 40
## 8   ./data/Mental-Rotation-2D_sub-001_date-20260924-095332.csv 40
## 9   ./data/Mental-Rotation-2D_sub-001_date-20260924-101256.csv 40
## 10  ./data/Mental-Rotation-2D_sub-001_date-20260926-211937.csv 40
## 11  ./data/Mental-Rotation-2D_sub-001_date-20260927-161502.csv 40
## 12  ./data/Mental-Rotation-2D_sub-001_date-20260927-181811.csv 40
## 13  ./data/Mental-Rotation-2D_sub-020_date-20260926-162037.csv 40
## 14 ./data/Mental-Rotation-2D_sub-1612_date-20260930-140849.csv 40
## 15                                              ./data/res.csv 40
## 16                                             ./data/res2.csv 40
```

Let's remove the file with less than 40 datapoints

```
files_ok <- setdiff(files, "Mental-Rotation-2D_sub-001_date-20260924-064330.csv")
```

```
# we recreate the data frame with the new list of files
data <- read_all(files_ok)
head(data)
```

```
##                                                  file_name subject_id
## 1 ./data/Mental-Rotation-2D_sub-000_date-20260924-094820.csv          0
## 2 ./data/Mental-Rotation-2D_sub-000_date-20260924-094820.csv          0
## 3 ./data/Mental-Rotation-2D_sub-000_date-20260924-094820.csv          0
## 4 ./data/Mental-Rotation-2D_sub-000_date-20260924-094820.csv          0
## 5 ./data/Mental-Rotation-2D_sub-000_date-20260924-094820.csv          0
## 6 ./data/Mental-Rotation-2D_sub-000_date-20260924-094820.csv          0
##   trial_idx angle condition response is_correct   rt
## 1         1    40  mirrored mirrored       TRUE 1649
## 2         2   120      same     same       TRUE 1433
```

```
## 3         3   160      same     same       TRUE 1168
## 4         4    80      same     same       TRUE  906
## 5         5    80  mirrored mirrored       TRUE  930
## 6         6   120      same     same       TRUE  800
```

Associating a unique subject_id to each file_name

```
data <- data %>%
  mutate(subject_id = factor(match(file_name, unique(file_name))))
head(data)
```

```
##                                                 file_name subject_id
## 1 ./data/Mental-Rotation-2D_sub-000_date-20260924-094820.csv          1
## 2 ./data/Mental-Rotation-2D_sub-000_date-20260924-094820.csv          1
## 3 ./data/Mental-Rotation-2D_sub-000_date-20260924-094820.csv          1
## 4 ./data/Mental-Rotation-2D_sub-000_date-20260924-094820.csv          1
## 5 ./data/Mental-Rotation-2D_sub-000_date-20260924-094820.csv          1
## 6 ./data/Mental-Rotation-2D_sub-000_date-20260924-094820.csv          1
##   trial_idx angle condition response is_correct   rt
## 1         1    40  mirrored mirrored       TRUE 1649
## 2         2   120      same     same       TRUE 1433
## 3         3   160      same     same       TRUE 1168
## 4         4    80      same     same       TRUE  906
## 5         5    80  mirrored mirrored       TRUE  930
## 6         6   120      same     same       TRUE  800
```

Error rates

Number of correct responses (out of 40) per subject:

```
data %>%
  group_by(subject_id) %>%
  summarise(n_correct = mean(is_correct) * 40)
```

```
## # A tibble: 16 x 2
##    subject_id n_correct
##    <fct>          <dbl>
##  1 1               39
##  2 2               38
##  3 3               38
##  4 4               30.8
##  5 5               39
##  6 6               39
##  7 7               37
##  8 8               39
##  9 9               39
## 10 10              40
## 11 11              40
## 12 12              40
## 13 13              40
## 14 14              40
## 15 15              35
## 16 16              39
```

Let's inspect the errors

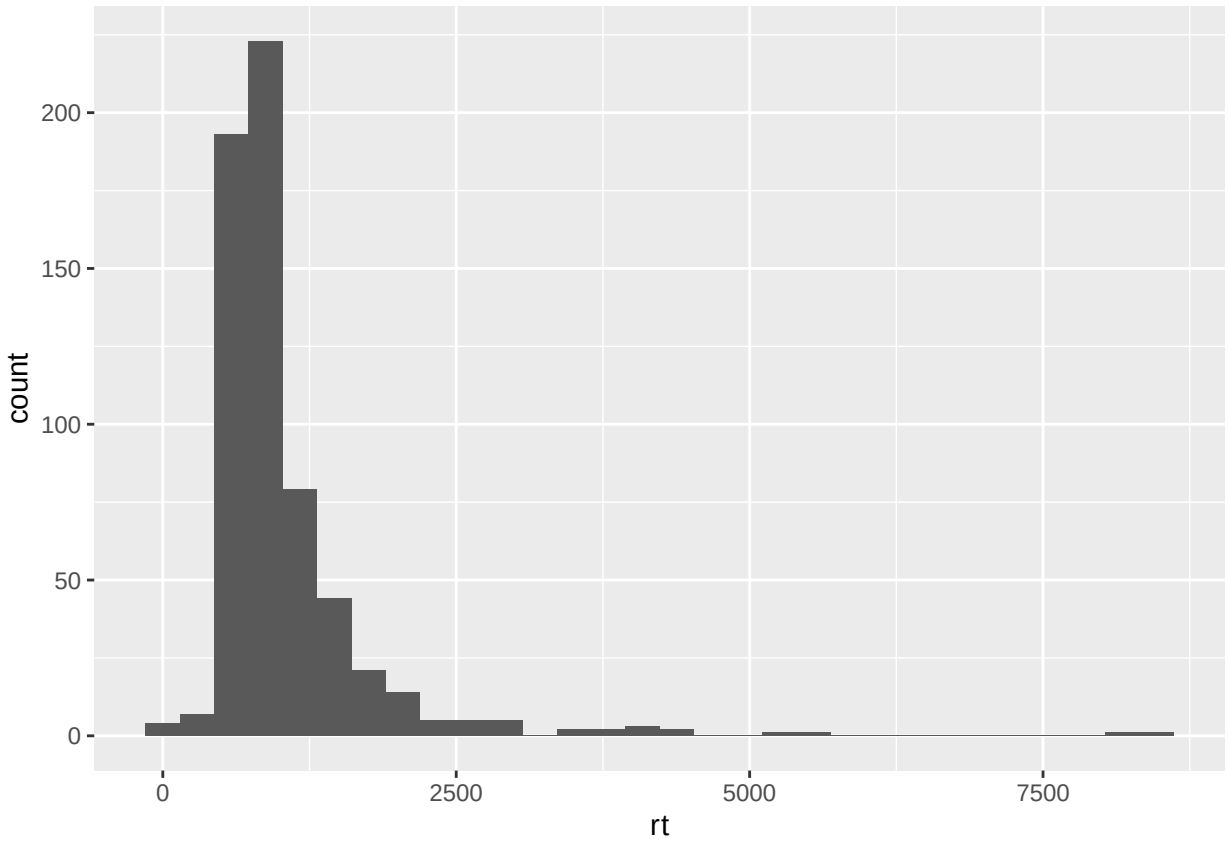
```
data %>% filter(!is_correct)
```

```
##                                                  file_name subject_id
## 1  ./data/Mental-Rotation-2D_sub-000_date-20260924-094820.csv          1
## 2  ./data/Mental-Rotation-2D_sub-000_date-20260924-095043.csv          2
## 3  ./data/Mental-Rotation-2D_sub-000_date-20260924-095043.csv          2
## 4  ./data/Mental-Rotation-2D_sub-000_date-20260924-095219.csv          3
## 5  ./data/Mental-Rotation-2D_sub-000_date-20260924-095219.csv          3
## 6  ./data/Mental-Rotation-2D_sub-001_date-20260924-064330.csv          4
## 7  ./data/Mental-Rotation-2D_sub-001_date-20260924-064330.csv          4
## 8  ./data/Mental-Rotation-2D_sub-001_date-20260924-064330.csv          4
## 9  ./data/Mental-Rotation-2D_sub-001_date-20260924-095036.csv          5
## 10 ./data/Mental-Rotation-2D_sub-001_date-20260924-095059.csv          6
## 11 ./data/Mental-Rotation-2D_sub-001_date-20260924-095218.csv          7
## 12 ./data/Mental-Rotation-2D_sub-001_date-20260924-095218.csv          7
## 13 ./data/Mental-Rotation-2D_sub-001_date-20260924-095218.csv          7
## 14 ./data/Mental-Rotation-2D_sub-001_date-20260924-095332.csv          8
## 15 ./data/Mental-Rotation-2D_sub-001_date-20260924-101256.csv          9
## 16                                            ./data/res.csv         15
## 17                                            ./data/res.csv         15
## 18                                            ./data/res.csv         15
## 19                                            ./data/res.csv         15
## 20                                            ./data/res.csv         15
## 21                                           ./data/res2.csv         16
##    trial_idx angle condition response is_correct   rt
## 1         13   120  mirrored     same      FALSE  984
## 2          1   120  mirrored     same      FALSE 1121
## 3         37   160      same mirrored      FALSE  550
## 4          8   120  mirrored     same      FALSE 1424
## 5         12    40      same mirrored      FALSE  932
## 6          4    80      same mirrored      FALSE 4184
## 7         11     0      same mirrored      FALSE  776
## 8         13    40  mirrored     same      FALSE  533
## 9         10   120  mirrored     same      FALSE 2057
## 10        30     0  mirrored     same      FALSE  639
## 11         5    40  mirrored     same      FALSE 1301
## 12         6    80      same mirrored      FALSE    0
## 13        35    80      same mirrored      FALSE  659
## 14         2   160      same mirrored      FALSE 1322
## 15        17   120      same mirrored      FALSE  544
## 16         4   160  mirrored     same      FALSE 1963
## 17         7   120      same mirrored      FALSE 1442
## 18        16    80  mirrored     same      FALSE 1412
## 19        19   120  mirrored     same      FALSE  725
## 20        38   160      same mirrored      FALSE 1340
## 21        23   160      same mirrored      FALSE 1428
```

Reaction times

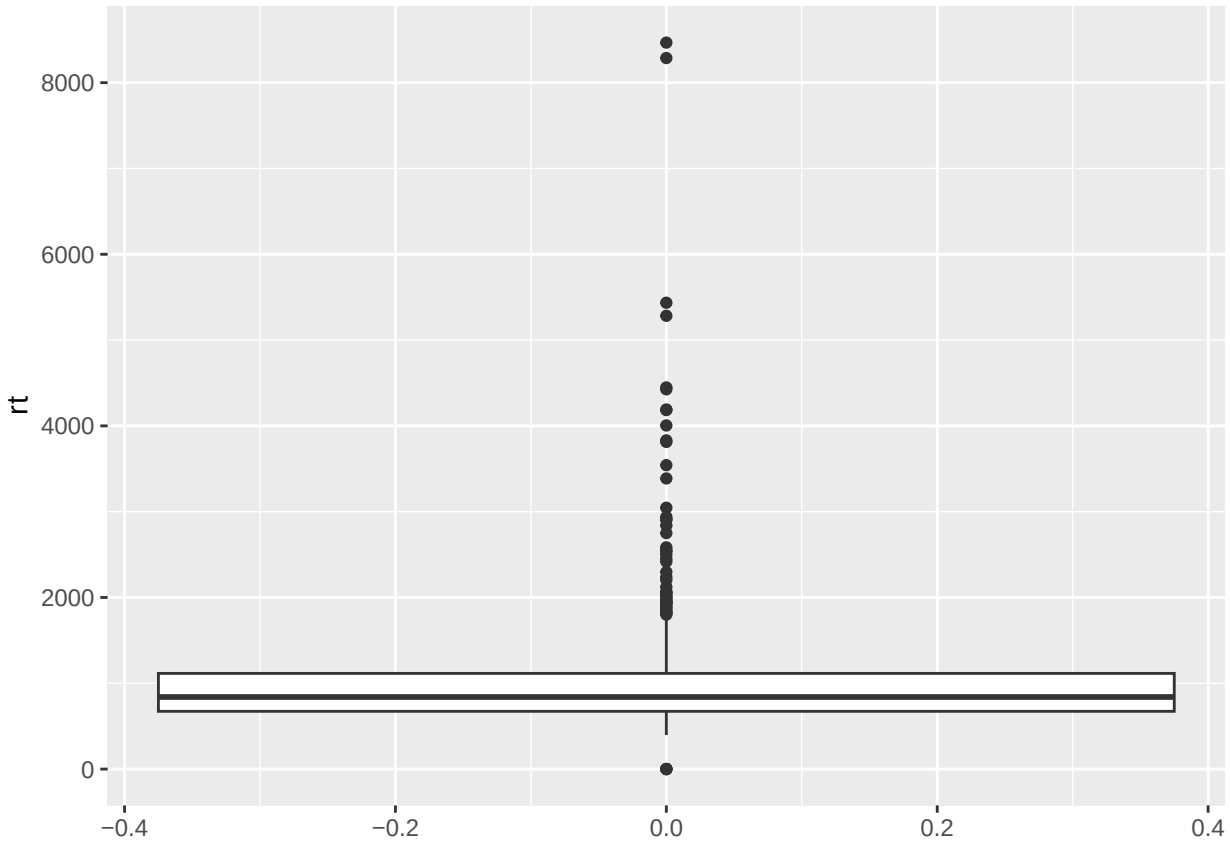
First, we inspect their distribution with an histogram

```
ggplot(data, aes(x = rt)) +
  geom_histogram(bins = 30)
```



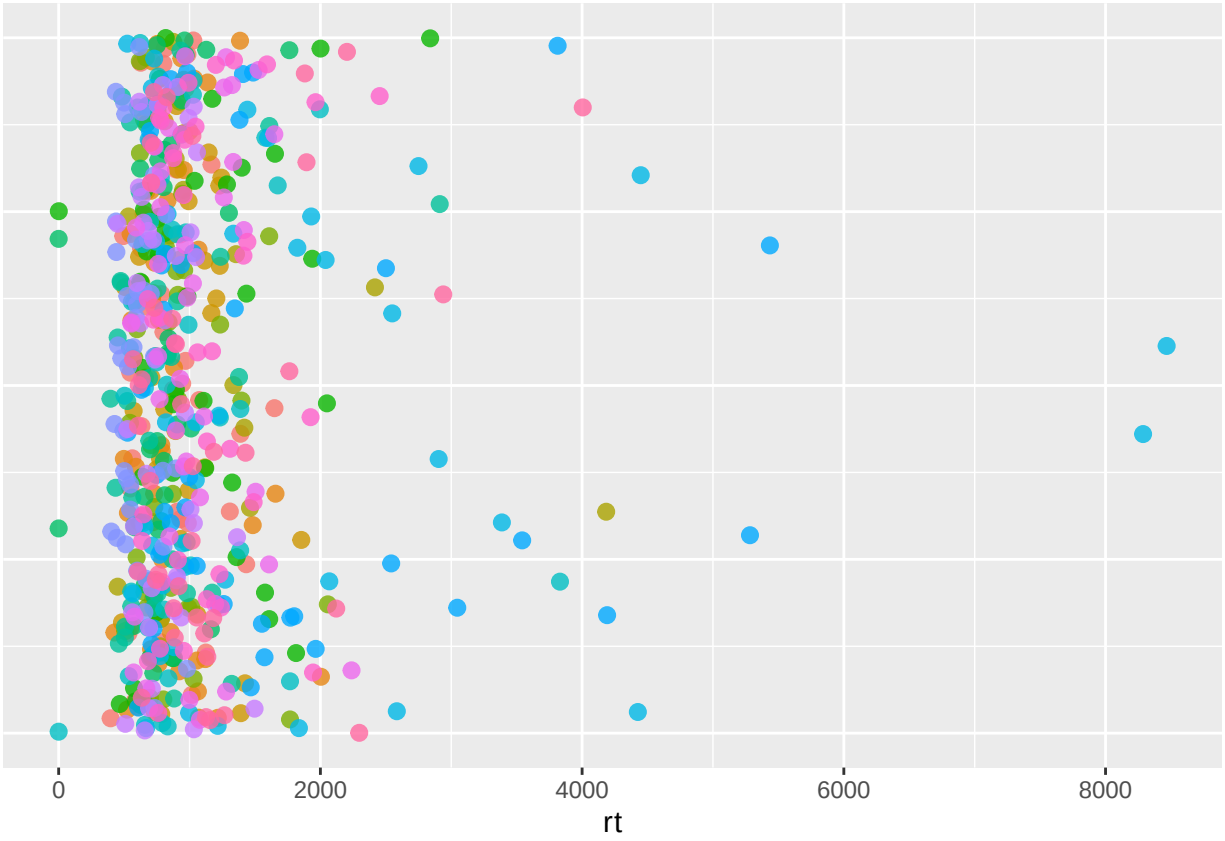
Then with a boxplot

```
ggplot(data, aes(y = rt)) +
  geom_boxplot()
```



A look at the individual differences

```
ggplot(data, aes(x = rt, y = 0, color = subject_id)) +
  # adds slight vertical spread so overlapping points remain visible
  geom_jitter(width = 0, height = 0.2, size = 2.5, alpha = 0.8) +
  theme(legend.position = "none",
        axis.title.y = element_blank(),
        axis.text.y = element_blank(),
        axis.ticks.y = element_blank())
```



Removing outliers

```
data_clean <- data %>% filter(rt > 150, is_correct)
```

```
print(nrow(data))                 # number of original data points
```

```
## [1] 613
```

```
nrow(data) - nrow(data_clean)     # number of rejected data points
```

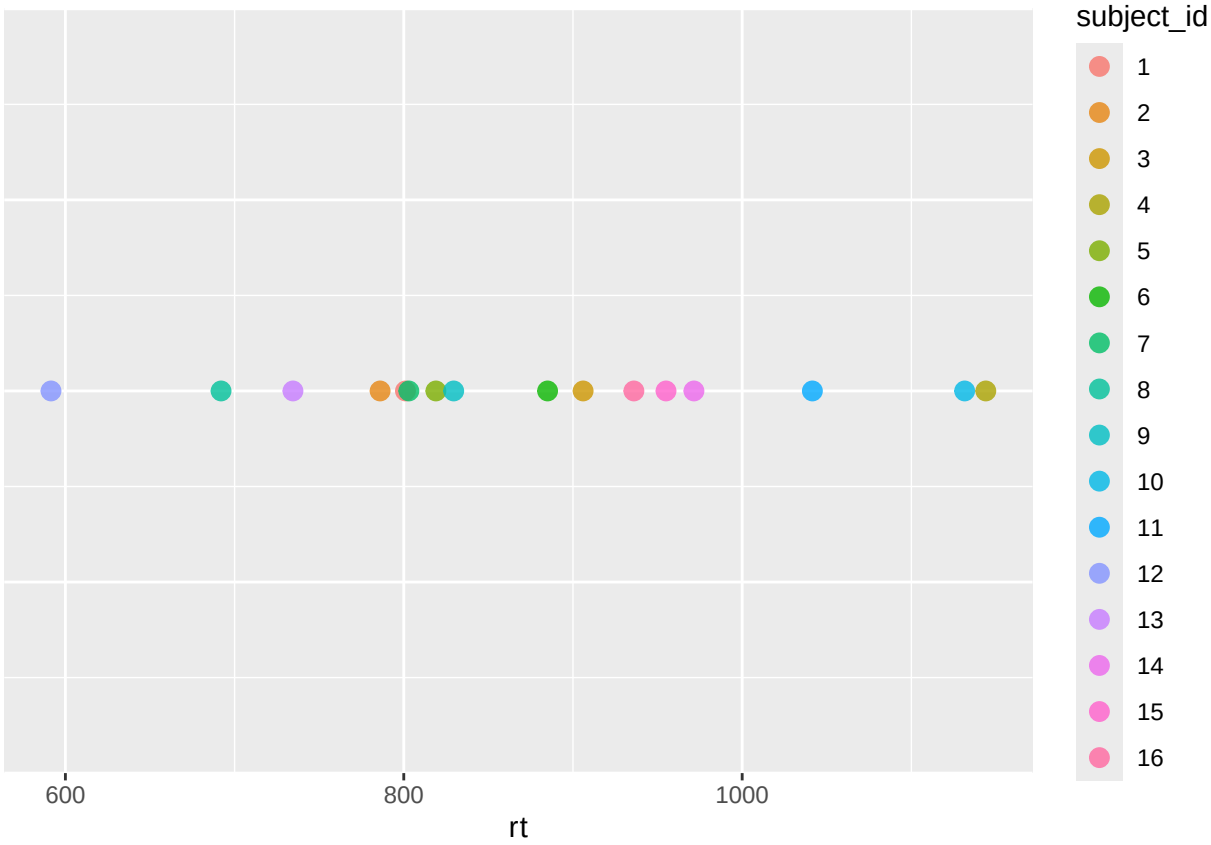
```
## [1] 24
```

```
medrt <- data_clean %>%
  group_by(subject_id) %>%
  summarise(rt = median(rt))
medrt
```

```
## # A tibble: 16 x 2
##    subject_id    rt
##    <fct>      <dbl>
##  1 1            801
##  2 2            786
##  3 3            906
##  4 4           1144
##  5 5            819
##  6 6            885
##  7 7            803
##  8 8            692
##  9 9            830.
```

```
## 10 10         1132.
## 11 11         1042.
## 12 12          592.
## 13 13          734.
## 14 14          972.
## 15 15          955
## 16 16          936
```

```
ggplot(medrt, aes(x = rt, y = 0, color = subject_id)) +
  geom_point(size = 3, alpha = 0.8) +
  theme(axis.title.y = element_blank(),
        axis.text.y = element_blank(),
        axis.ticks.y = element_blank())
```



Effects of angle and condition (same vs mirror)

```
rts2 <- data_clean %>%
  group_by(subject_id, condition, angle) %>%
  summarise(rt = median(rt), .groups = "drop")
rts2
```

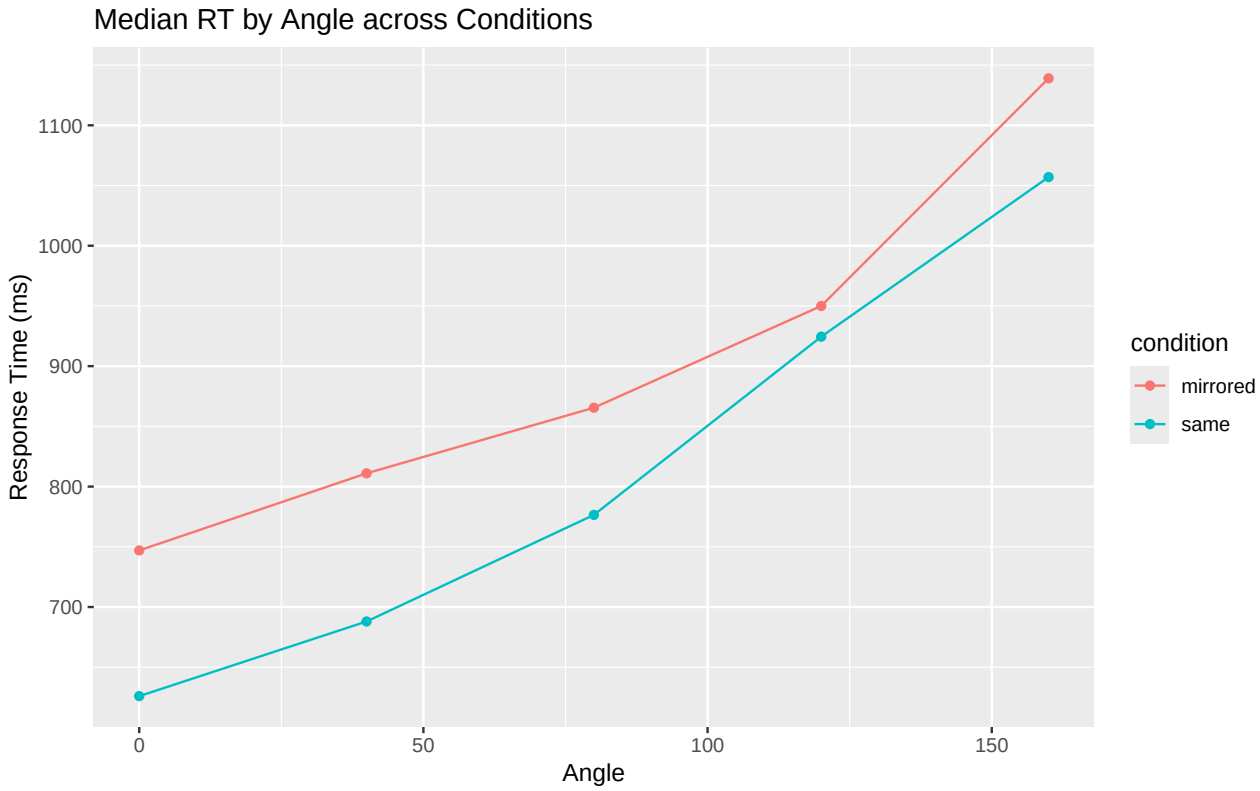
```
## # A tibble: 157 x 4
##    subject_id condition angle    rt
##    <fct>      <chr>     <int> <dbl>
##  1 1          mirrored      0  798
##  2 1          mirrored     40 1046
##  3 1          mirrored     80  870
```

```
##  4 1          mirrored    120 1029
##  5 1          mirrored    160  902.
##  6 1          same          0  572
##  7 1          same         40  548
##  8 1          same         80  814.
##  9 1          same        120  872
## 10 1          same        160 1011
## # i 147 more rows
```

```
rts3 <- data_clean %>%
  group_by(condition, angle) %>%
  summarise(rt = median(rt), .groups = "drop")
rts3
```

```
## # A tibble: 10 x 3
##    condition angle    rt
##    <chr>     <int> <dbl>
##  1 mirrored      0  747
##  2 mirrored     40  811
##  3 mirrored     80  866.
##  4 mirrored    120  950
##  5 mirrored    160 1139
##  6 same          0  626
##  7 same         40  688
##  8 same         80  776.
##  9 same        120  924.
## 10 same        160 1057
```

```
ggplot(rts3, aes(x = angle, y = rt, color = condition)) +
  geom_line() +
  geom_point() +  # adds points at each angle
  labs(x = "Angle", y = "Response Time (ms)",
       title = "Median RT by Angle across Conditions")
```

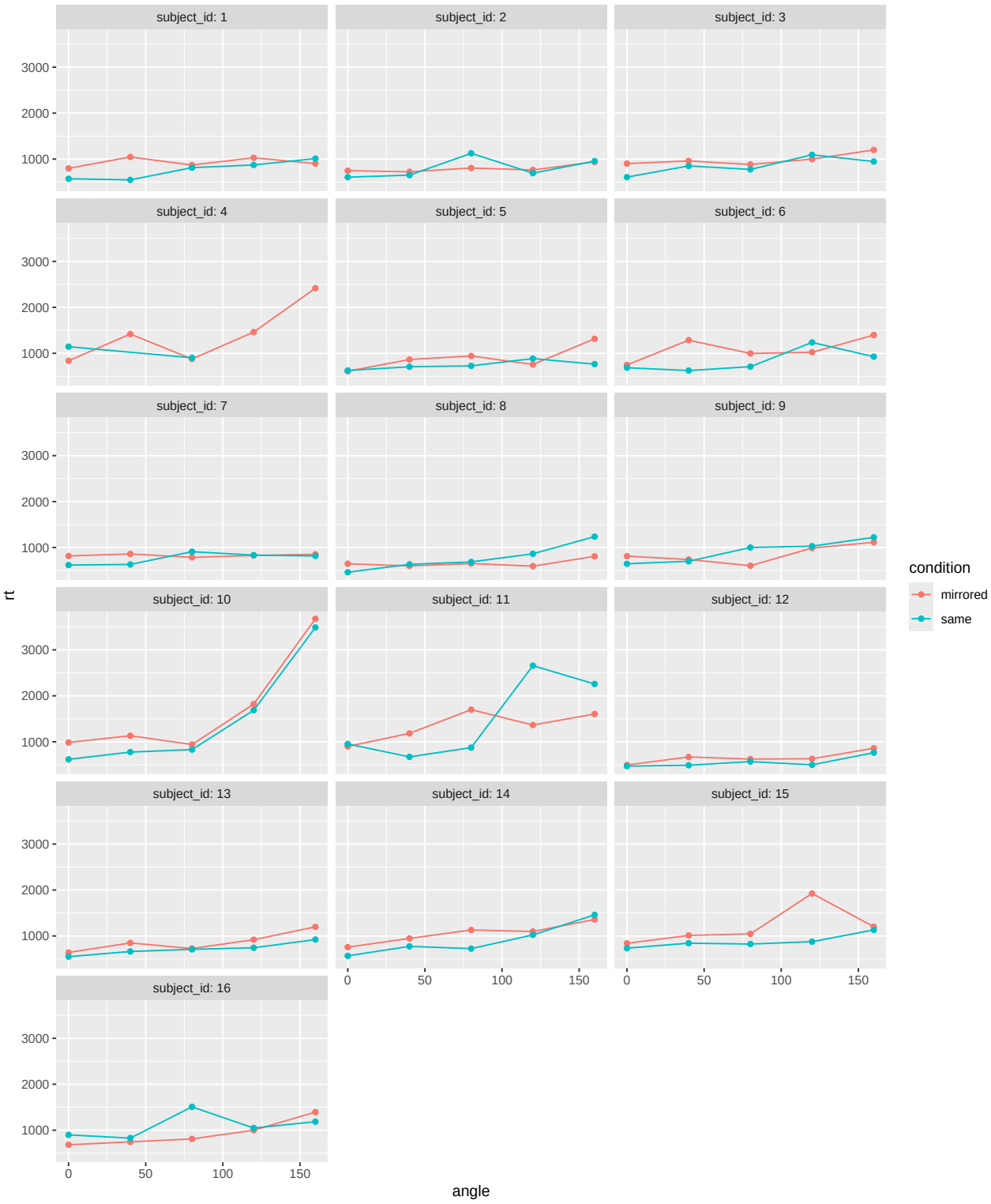


Inferential analysis

Are the differences statistically significant? That is, what is the probability that they are due to chance fluctuations?

If we find that the probability is small, then we will reject the "null" hypothesis. The approach is called "Hypothesis testing".

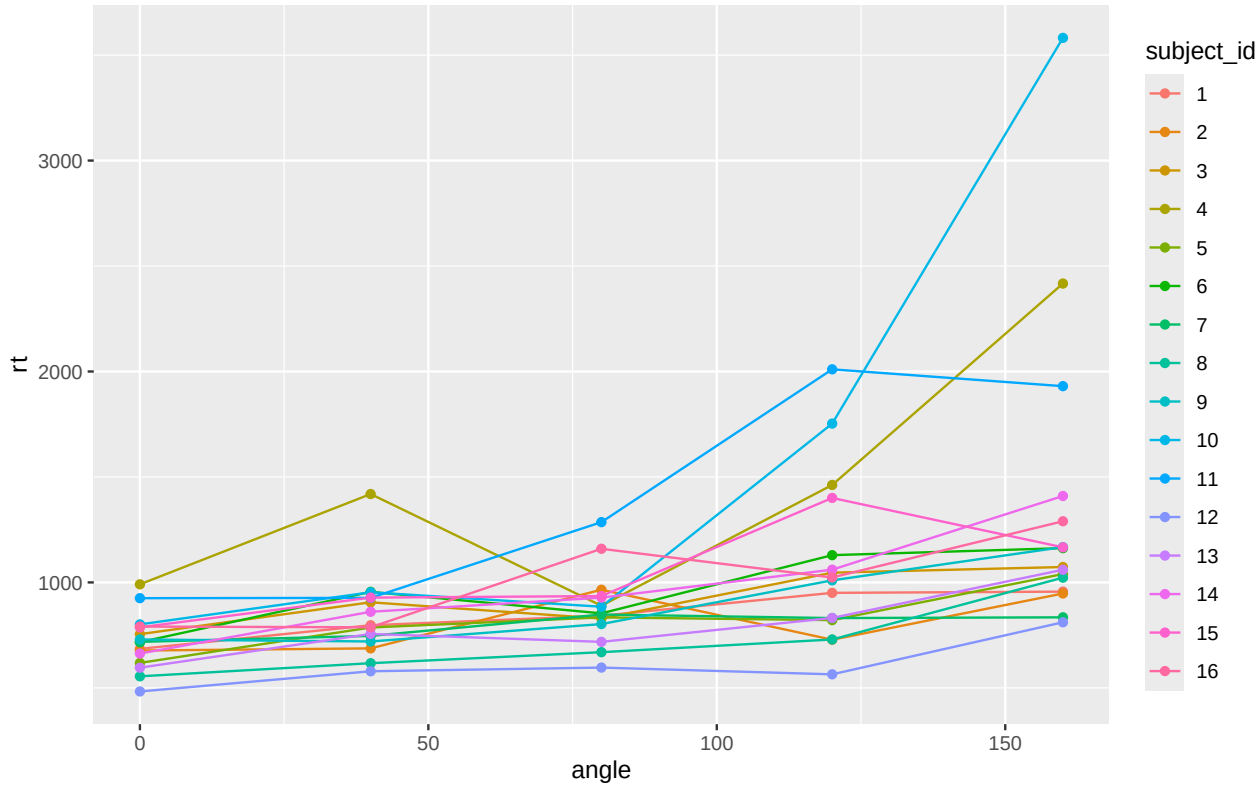
```
ggplot(rts2, aes(x = angle, y = rt, color = condition)) +
  geom_line() +
  geom_point() +
  facet_wrap(~ subject_id, ncol = 3, labeller = label_both)  # one subplot per subject
```



Let's forget the distinction between mirror & same

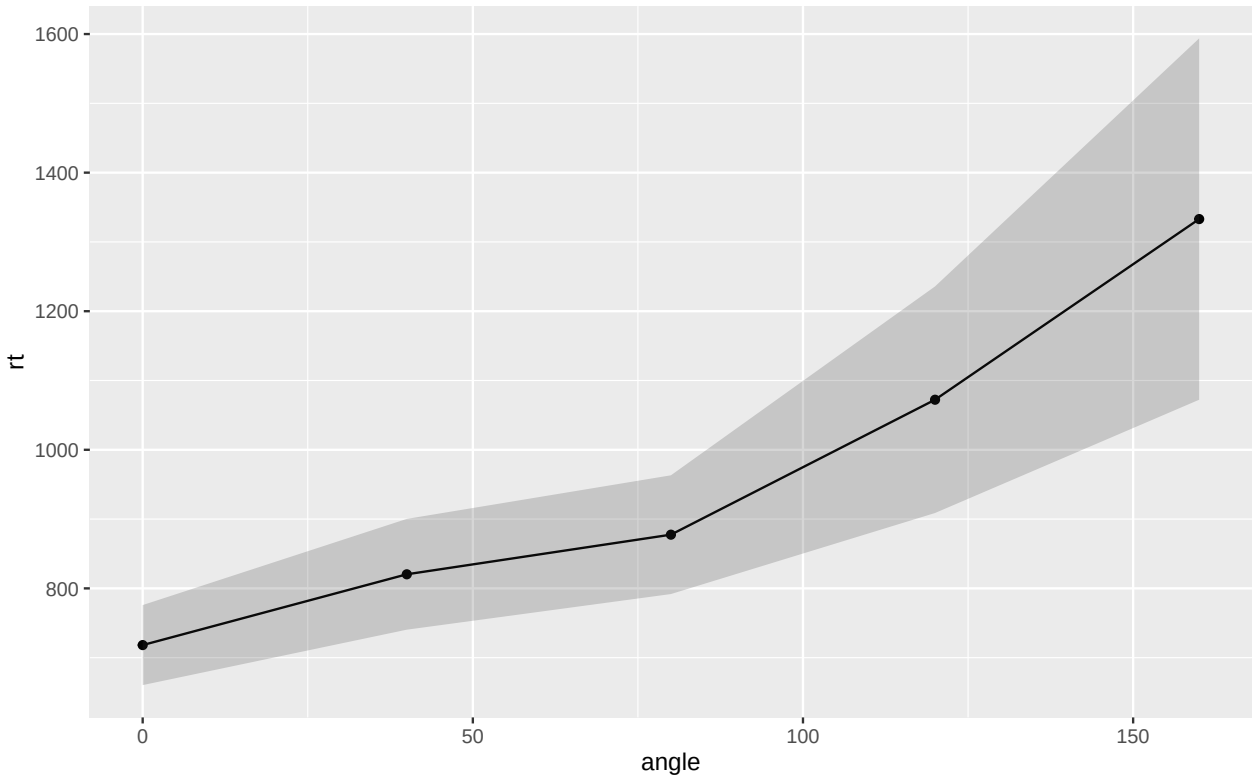
```
# individual results superposed (averaging over condition)
rts2 %>%
  group_by(subject_id, angle) %>%
  summarise(rt = mean(rt), .groups = "drop") %>%
```

```
  ggplot(aes(x = angle, y = rt, color = subject_id)) +
  geom_line() +
  geom_point()
```




```
# group result, with 95% confidence intervals (t-based)
mean_ci95 <- function(x) {
  ci <- t.test(x)$conf.int
  data.frame(y = mean(x), ymin = ci[1], ymax = ci[2])
}

ggplot(rts2, aes(x = angle, y = rt)) +
  stat_summary(fun = mean, geom = "line") +
  stat_summary(fun = mean, geom = "point") +
  stat_summary(fun.data = mean_ci95, geom = "ribbon", alpha = 0.2)
```



Let's compute p-values for the differences between angles conditions

```
angles_rt <- rts2 %>%
  group_by(angle, subject_id) %>%
  summarise(rt = mean(rt), .groups = "drop")
angles_rt
```

```
## # A tibble: 80 x 3
##    angle subject_id    rt
##    <int> <fct>      <dbl>
##  1     0 1           685
##  2     0 2           677
##  3     0 3           755.
##  4     0 4           991
##  5     0 5           618.
##  6     0 6           717.
##  7     0 7           718.
##  8     0 8           555.
##  9     0 9           730.
## 10     0 10          801
## # i 70 more rows
```

```
# rows are sorted by subject_id within each angle, so the vectors are paired
rt0   <- angles_rt %>% filter(angle == 0)   %>% pull(rt)
rt40  <- angles_rt %>% filter(angle == 40)  %>% pull(rt)
rt80  <- angles_rt %>% filter(angle == 80)  %>% pull(rt)
rt120 <- angles_rt %>% filter(angle == 120) %>% pull(rt)
rt160 <- angles_rt %>% filter(angle == 160) %>% pull(rt)
```

```
rt40 - rt0
```

```
##  [1] 112.00  10.25 151.00 428.00 168.00 238.50  28.25  62.25  -9.50 152.25
## [11]   1.50  95.75 160.00 198.00 141.00  -3.50
```

Let's compute the p-values using a permutation test (see https://en.wikipedia.org/wiki/Permutation_test)

```
my_permutation_test <- function(sample, npermutations = 10000) {
  # test if sample mean is significantly above 0 (one-tail test)
  obs_mean <- mean(sample)
  p <- length(sample)
  h0_dist <- replicate(npermutations,
                       mean(sample * sample(c(1, -1), size = p, replace = TRUE)))
  mean(h0_dist >= obs_mean)  # p-value
}
```

```
print(mean(rt40 - rt0))
```

```
## [1] 120.8594
```

```
my_permutation_test(rt40 - rt0)
```

```
## [1] 2e-04
```

```
my_permutation_test(rt80 - rt40)
```

```
## [1] 0.2405
```

```
my_permutation_test(rt120 - rt80)
```

```
## [1] 0.0068
```

```
my_permutation_test(rt160 - rt120)
```

```
## [1] 0.0026
```

Using R's built-in tests

Wilcoxon signed-rank test

The closest built-in equivalent of the sign-flipping permutation test is the Wilcoxon signed-rank test, which is a permutation test on the ranks of the differences (two-sided by default, like `scipy.stats.permutation_test`).

```
wilcox.test(rt40 - rt0, mu = 0)
```

```
##
##  Wilcoxon signed rank exact test
##
## data:  rt40 - rt0
## V = 131, p-value = 0.0003052
## alternative hypothesis: true location is not equal to 0
```

Student t-test

See https://en.wikipedia.org/wiki/Student%27s_t-test

```
# computing the t value "à la mano"
d <- rt40 - rt0
m <- mean(d)
s <- sd(d)      # R's sd() divides by n-1 (numpy's std() divides by n)
n <- length(d)
t <- m / s * sqrt(n)
print(c(m = m, s = s, n = n, t = t))
```

```
##          m          s          n          t
## 120.859375 113.089786  16.000000   4.274811
```

```r
t.test(d, mu = 0)
```

```
##
##  One Sample t-test
##
## data:  d
## t = 4.2748, df = 15, p-value = 0.0006647
## alternative hypothesis: true mean is not equal to 0
## 95 percent confidence interval:
##   60.59808 181.12067
## sample estimates:
## mean of x
##  120.8594
```